Pipeline Setup Checklist

Every value to obtain and every check to pass before the robot-inspection pipeline will run. Tick each box; write your value in the blank column.

1. Values to obtain (write them down)

Credentials & access:

Item	Where to get it / example	Your value
Spot IP address	From your Spot admin / on Spot's GXP: 192.168.50.3	
Spot username	Boston Dynamics account for the robot	
Spot password	Boston Dynamics account for the robot	
GitHub repo access	arunvaradarajan8/robot-inspection (login/token if private)	
Trimble Perspective license	Installed on the laptop, X7 paired	

Network values (see the network setup guide):

Item	Where to get it / example	Your value
Beryl Wi-Fi SSID	You choose, e.g. spot-field (5 GHz)	
Beryl Wi-Fi password	You choose (WPA2/WPA3)	
Beryl LAN subnet / gateway	10.0.0.0/24 gateway 10.0.0.1	
Jetson IP (on Beryl)	Static lease 10.0.0.10	
Laptop dongle IP (on Beryl)	Static lease 10.0.0.20	
Spot payload (GXP) IP	192.168.50.3 (confirm your payload config)	
X7 Wi-Fi SSID / password	The scanner's own AP (for Perspective)	
ROS_DOMAIN_ID	42 (Jetson only)	

2. config/field.env values to set / confirm

On the Jetson, in config/field.env. The localization block is the critical one for camera-only operation.

Item	Where to get it / example	Your value
SPOT_IP	Spot's address (192.168.50.3)	
SPOT_USERNAME / SPOT_PASSWORD	Spot login (blank = fill in)	
TRIMBLE_WINDOWS_URL	http://10.0.0.20:8765 (laptop dongle IP)	
ROS_DOMAIN_ID	42	
IMAGE_TOPIC	Must match your camera driver (ros2 topic list)	
MAP_DEPTH_POINTS_TOPIC	The camera's point cloud, e.g. /depth/points	
ROBOT_GOAL_BRIDGE	false for supervised (recommended first)	

Localization = camera only (do NOT change these):

Item	Where to get it / example	Your value
SPOT_LOCALIZATION	false (Spot's leg-encoder odometry OFF)	
DEPTH_LOCALIZATION	true (Oak-D VIO is the localizer)	
ROBOT_WORLD_FRAME	depth_odom	
DEPTH_ODOM_TOPIC	/oak/odom (must match your camera's VIO topic)	
FUSED_LOCALIZATION	false (on would re-add Spot's pose)	

CRITICAL: localization now comes only from the Oak-D. Your camera launch MUST publish visual-inertial odometry on /oak/odom (on-device VIO or an RTAB-Map/VIO node). No /oak/odom = no localization = no map.

3. Jetson - install & verify

✓	What to check	How
☐	Ubuntu 24.04 / JetPack 6	cat /etc/nv_tegra_release
☐	Repo cloned to ~/ros2_ws	ls ~/ros2_ws (src scripts config docs tools)
☐	ROS 2 Jazzy installed	ros2 --help
☐	Python libs	pip3 show numpy opencv-python pyyaml
☐	Spot SDK + scan libs	python3 -c 'import bosdyn.client, laspy'
☐	Depth-camera driver (depthai-ros)	ros2 pkg prefix depthai_ros_driver
☐	Workspace builds	cd ~/ros2_ws && colcon build --symlink-install
☐	Camera point cloud is live	ros2 topic hz /depth/points
☐	Camera VIO odometry is live	ros2 topic hz /oak/odom (localization!)
☐	Topic names match field.env	ros2 topic list | grep -E 'rgb|depth|points|odom'
☐	No-hardware sim runs	./scripts/run_field.sh mission --sim synthetic

4. Windows laptop - install & verify

✓	What to check	How
☐	Trimble Perspective scans the X7 on its own	Take a test scan first
☐	Python 3.12 installed (Added to PATH)	py --version
☐	Repo downloaded (ZIP or git clone)	robot-inspection folder present
☐	Bridge deps installed	Run 'Install Windows Dependencies.bat'
☐	Bridge app launches	py windows_app.py -> dashboard on :8765
☐	Bridge settings filled	jetson_host 10.0.0.10, windows_advertise_host 10.0.0.20
☐	export_dir points at Perspective's export folder	Settings tab

✓	What to check	How
☐	auto_transfer is OFF	Settings tab
☐	Firewall allows inbound 8765	New-NetFirewallRule ... -LocalPort 8765
☐	USB Wi-Fi adapter on the Beryl	Built-in Wi-Fi stays on the X7

5. Hardware & wiring

✓	What to check	How
☐	Jetson Orin Nano has Wi-Fi (M.2 AX210)	nmcli device status -> wlan0
☐	Jetson Ethernet wired to Spot GXP	ping 192.168.50.3 from Jetson
☐	Oak-D Pro connected + udev rule set	USB 3; 80-movidius.rules
☐	Camera->body transform (extrinsic) published	ros2 run tf2_tools view_frames
☐	Beryl powered, LAN 10.0.0.0/24, AP isolation OFF	admin at 10.0.0.1
☐	Static leases set (Jetson .10, laptop .20)	Beryl Address Reservation
☐	Spot powered, tablet + E-Stop in hand	
☐	Trimble X7 + tripod ready, paired in Perspective	

6. Network connectivity - test both directions

✓	What to check	How
☐	Jetson reaches the laptop bridge	curl http://10.0.0.20:8765/status
☐	Jetson reaches Spot	ping 192.168.50.3
☐	Laptop can SSH the Jetson	ssh <user>@10.0.0.10 "echo ok"
☐	Perspective still sees the X7	On the laptop's X7 Wi-Fi
☐	Preflight passes	./scripts/field_preflight.sh mission

7. First run - supervised (no autonomous driving)

✓	What to check	How
☐	Perspective + bridge app running on the laptop	
☐	Mission starts with motion OFF	ROBOT_GOAL_BRIDGE=false ./scripts/run_field.sh mission
☐	Map grows as you walk Spot by tablet	ros2 topic echo /mission/state
☐	Robot parks at a scan spot and WAITS	expected - operator paced
☐	'Next scan location' releases it to the next	press in the bridge app after scanning
☐	Returns to start -> AWAITING_UPLOAD	ros2 topic echo /mission/state
☐	Upload E57 + Finish closes the mission	bridge app button

Autonomous driving (ROBOT_GOAL_BRIDGE=true + spot_sdk) works with camera-only localization: the goal bridge makes each goal body-relative from the camera TF, then phrases it in Spot's frame from Spot's snapshot (motion dialect only - never localization). Still verify supervised runs first, E-Stop in hand.

8. Optional - tuning values to decide

Item	Where to get it / example	Your value
EXPLORATION_RADIUS_M	How far to map (default 40)	
MISSION_MAX_EXCURSION_M	>= exploration radius (default 40)	
SCAN_MAX_STATIONS	How many scans to take (default 3)	
SCAN_MIN_SEPARATION_M	Spacing between scans (default 8)	
SCAN_MIN_OPENNESS / SCAN_OPENNESS_RADIUS_M	Vantage clearance (0.6 / 3.0)	
SCAN_CENTRALITY_SCALE_M	Bias toward structure centre (default 10)	
PLANNER_TRAVEL_DECAY_M	Greedy vs. reach (default 8)	